

# Amazon Sales Analysis

Professional Data Analytics Case Study – Python · MySQL · Power BI · GitHub

DATA ANALYST PORTFOLIO



# Business Problem

Organizations need a **centralized dashboard** to monitor sales, customer behavior, product performance, payment methods, and regional sales trends – enabling better, data-driven decision-making across the business.



# Dataset Overview

|     | Ne Meesadās  | d | Expiads  | 0        | Fag    | Newn   | Tho       | San tāao micrucs |                | Sa         | Bo            | Mler  | Do    | Fne     |               |
|-----|--------------|---|----------|----------|--------|--------|-----------|------------------|----------------|------------|---------------|-------|-------|---------|---------------|
|     | S093maist    | A | 1, 0300  | (r-7/1%) | \$023  | \$00   | (% (avls) | 5/% (s:we)       | Midssas (ante) | 503 (ssme) | Sh10 (s:ssit) | J9    | 3,96  | 2336    | (so na mnite) |
| 1   | S000runse    |   | \$790    |          | \$,000 | 14316  | -         | 16,03            |                | 05         | 366           | 2,76  | 3,56  | 3,336   |               |
| 2   | 5060mes      |   | \$1,00   |          | C6,34  | 11393  | -         | \$1933           | 160            | \$56       | 328           | 9,33  | 5,63  | \$6,106 |               |
| 3   | 2010mnds     |   | \$190    |          | C3,06  | 11633  | -         | \$1933           |                | 5,45       | 396           | 7,38  | 3,96  | \$,386  |               |
| 3   | 2589mes      |   | \$500    |          | 1K5,33 | \$,966 | -         | \$3600           |                | \$,09      | 180           | 3,46  | 56    | \$,816  |               |
| 4   | 4039mnes     |   | \$6105   |          | 34016  | 5,238  | -         | \$5953           |                | \$,06      | 118           | 3,96  | 1,48  | \$196   |               |
| 5   | 5989nnus     |   | \$5356   |          | 5%,00  | \$,258 | nuv       | \$77121          | \$,90          | \$,58      | 1,58          | \$,56 | 83    | \$2,016 |               |
| 7   | C010roes     |   | \$26208  |          | 14,00  | \$3000 | -         | \$60192          | \$,38          | \$3,36     | 89            | 18    | 1,33  | 33,96   |               |
| 6   | 0059mme      |   | \$1790   |          | 131,54 | \$1158 | -         | \$2,03           |                | \$,60      | 190           | 58    | 9,40  | 69,53   |               |
| 5   | L090roee     |   | \$7,00   |          | 34,04  | 3,423  | -         | \$6,33           |                | \$,90      | 53            | 420   | 60    | 3,08    |               |
| 6   | 0026mee      |   | 1\$8,101 |          | 16,56  | 16343  | -         | \$5,323          |                | \$,23      | 53            | 63    | 36    | 5,63    |               |
| 8   | S980uudân    |   | \$9,00   |          | \$462  | 1,336  | -         | \$1300           | \$50           | 143        | 00            | \$20  | 1,60  | 3,408   |               |
| 17  | 3928mns      |   | \$3,20   |          | 68,04  | 51930  | -         | \$1220           | \$90           | 141        | 36            | 15    | 19    | \$125   |               |
| 13  | 3390mes      |   | \$4,06   |          | 17238  | 18390  | -         | \$7,232          |                | \$90       | 18            | 23    | 1,65  | \$8,186 |               |
| 16  | S000mines    |   | \$0,00   |          | 85,00  | 11800  | -         | \$3,330,838      |                | \$90       | 57            | 41    | 20    | \$1,836 |               |
| 17  | S020mues     |   | \$9,06   |          | 18,03  | \$1433 | -         | \$3299           |                | \$00       | 43            | 35    | 1,23  | 16,688  |               |
| 15  | S249mus      |   | \$170    |          | 18,04  | \$1990 | -         | \$0893           |                | \$06       | 560           | 49    | 1,23  | 3,035   |               |
| 16  | S4151me      |   | \$7,96   |          | \$8,00 | \$1,41 | -         | \$5,025          |                | \$31       | 40            | 45    | \$,36 | 7,16    |               |
| 13  | S496.0allenb |   | 35,06    |          | 14,98  | 01195  | -         | \$3,595          |                | 457        | 20            | 55    | 150   | 7,16    |               |
| 15  | Gaencees     |   | 146,00   |          | 12068  | \$8500 | -         | \$996,830        |                | 4,30       | 110           | 25    | 680   | 7,16    |               |
| 10  |              |   | \$6,08   |          | 16960  | \$3308 | -         | \$5,232\$30      |                | \$,36      | 110           | 83    | 1,16  | 7,16    |               |
| 110 |              |   | \$8,011  |          | 16033  | 11,00  |           | \$3,183          |                | \$,50      | 71            | 0,60  |       | 7,16    |               |

## Amazon Sales Dataset

Key fields: Order Date, Customer, Product, Category, Brand, Quantity, Unit Price, Discount, Shipping Cost, Total Sales, Payment Method, State, City.

10,000  
Rows

21  
Columns



# Data Cleaning with Python

Raw transactional data was cleaned and validated using Python before SQL analysis.

## → Remove Duplicates

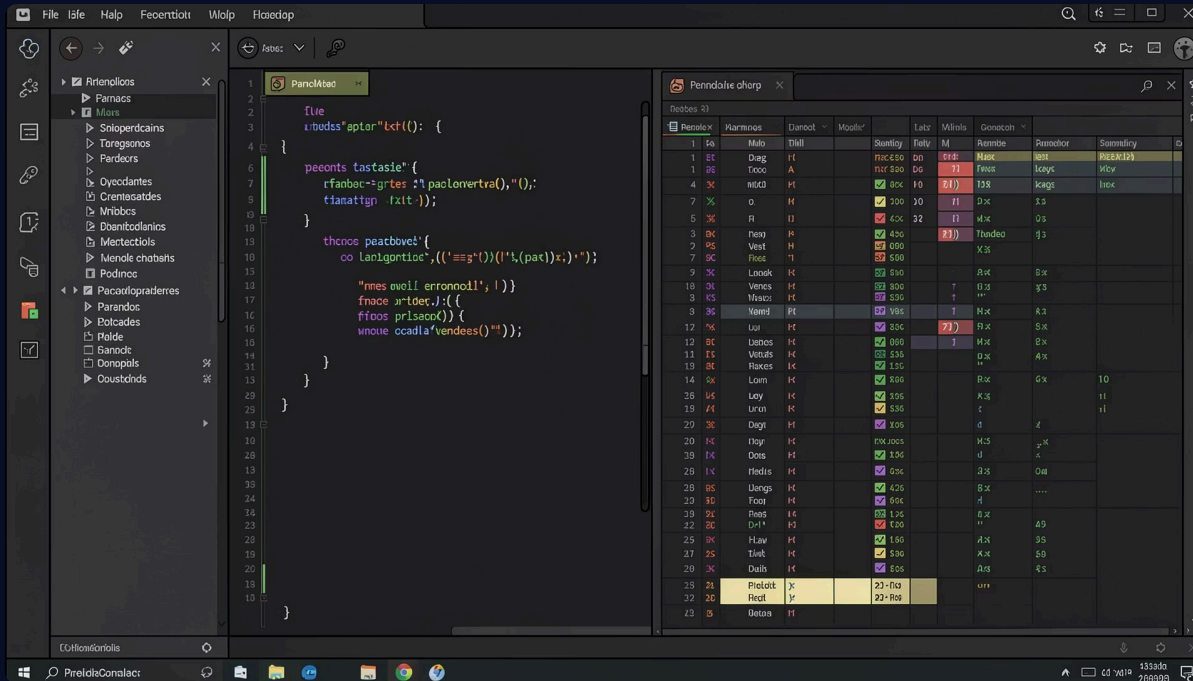
Eliminated duplicate records from the dataset.

## → Handle Missing Values

Checked and addressed null entries.

## → Standardize & Validate

Standardized column names, converted dates, validated numeric columns, and exported the cleaned dataset for SQL analysis.



# SQL Business Analysis

The SQL phase answered critical business questions across sales, customers, and operations.



## Sales Metrics

Total Sales, Total Orders, Average Order Value, Monthly Sales Trend



## Product Performance

Sales by Category, Sales by Brand, Top Products



## Customer & Regional

Top Customers, Sales by State



## Payment & Orders

Payment Method Analysis, Order Status Analysis

# Power BI Dashboard



## What's Inside

The interactive dashboard consolidates all key metrics into a single view, enabling stakeholders to explore sales trends, category performance, geographic distribution, and payment behavior at a glance.

# Key Business Insights

## Category Performance

Sales performance varies significantly across product categories.

## Product Demand Trends

Product demand changes over time.

## Brand Revenue Contribution

Certain brands contribute more revenue than others.

## Geographic Hotspots

Geographic analysis highlights high-performing regions.

## Customer Behavior

Payment method preferences provide customer behavior insights.

# Business Recommendations

Actionable strategies derived from the analysis to drive growth and efficiency.



## Invest in Top Categories

Increase investment in top-performing categories.



## Optimize Inventory

Optimize inventory for high-demand products.



## Expand Regional Marketing

Expand marketing in high-performing regions.



## Improve Retention

Improve customer retention strategies.



## Track KPIs

Track KPIs through interactive dashboards.



# Skills & GitHub Repository

## Skills Demonstrated

Python (Pandas)

Power BI & DAX

Business Analysis

MySQL & SQL

Data Cleaning &  
Visualization

Git & GitHub

## Repository Structure

Amazon-Sales-Analysis/

├── data/

├── sql/

├── dashboard/

├── images/

├── README.md

└── LICENSE

# Conclusion

This project demonstrates the complete data analytics lifecycle – from raw data preparation to business intelligence dashboards and actionable recommendations.

Designed to showcase practical skills expected from an entry-level **Data Analyst**, this end-to-end solution covers data cleaning, SQL-based business analysis, Power BI visualization, and business recommendations.

## Python

Data Cleaning

## MySQL

Business Analysis

## Power BI

Dashboard

## GitHub

Portfolio

